

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Executed laboratory data review, trending, and reporting workflows using **Python** (pandas, NumPy) and **Power BI** to collate, analyse, and visualize end-of-quarter and year-end batch data for regulatory compliance and operational decision-making.
- Supported equipment qualification (PQ) documentation generation and standard operating procedure (SOP) development for enterprise laboratory systems, ensuring **GMP** alignment and **USP/EP/JP** regulatory requirements across **LIMS** and electronic laboratory notebooks.
- Developed and maintained **Process Analytical Technology (PAT)** monitoring dashboards and data pipelines using **SQL** (PostgreSQL, MySQL, Oracle) and **Python** scripting to automate laboratory troubleshooting, method verification execution, and facility testing workflows; reduced manual data processing by **40%**.
- Managed laboratory consumables ordering, stock maintenance, and facility documentation using enterprise business systems integration via **REST APIs** and **LIMS** platforms; coordinated with cross-functional operations and microbiology lab teams to ensure continuous improvement in laboratory processes and housekeeping standards.
- Participated in internal and external audits, functional meetings, and cross-functional facilitation sessions to support site safety culture, PAT adoption, and introduction of novel biologics testing solutions; escalated non-routine partner and customer requests to management with documented audit trails.
- Coached and supported team members on laboratory systems (E-logs, MODA, electronic notebooks), training completion, and process documentation; demonstrated flexibility in supporting exceptional out-of-hours facility operations and changing business needs within a microbiology-focused laboratory environment.

SKILLS

Laboratory & Process Systems – LIMS, E-logs, MODA, electronic notebooks, Process Analytical Technology (PAT), equipment qualification (PQ), SOP development, GMP, USP, EP, JP regulatory requirements.

Data Analysis & Reporting – Power BI, Tableau, Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), data modelling, ETL, batch data analysis, trend reporting.

Programming & Automation – Python, Java, SQL, REST APIs, Flask, FastAPI, shell scripting, Linux, Git, Jenkins CI/CD.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, containerization, system monitoring and alerting.

Biologics & Microbiology Domain – microbiological testing techniques, biologics manufacturing, facility operations, method verification, troubleshooting, change-control management.

Operational & Team Skills – cross-functional collaboration, audit support, continuous improvement, safety culture, team coaching, documentation, Jira, Confluence, Agile/Scrum.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Laboratory Data Pipeline & PAT Monitoring System (HCL Technologies)

- Architected and deployed an end-to-end laboratory data pipeline using **Python**, **PostgreSQL**, and **Power BI** to automate collection, validation, and reporting of facility testing data, method verification results, and batch-level troubleshooting outcomes for regulatory submission and GMP audits.
- Implemented **LIMS** and electronic notebook integrations via **REST APIs** to harmonize raw laboratory data across multiple equipment qualification (PQ) systems; established automated UAT coordination and post-go-live support workflows that reduced data reconciliation time by **35%**.
- Developed **Process Analytical Technology (PAT)** dashboards with dynamic **DAX** measures and **Power Query** ETL to support real-time facility testing visibility, trend analysis for continuous improvement, and compliance reporting against **USP/EP/JP** and **GMP** standards for microbiology lab operations.

SOP Documentation & Laboratory Operations Automation (HCL Technologies)

- Led documentation generation of standard operating procedures (SOPs) and work instructions for equipment qualification (PQ), laboratory housekeeping, consumables management, and facility testing protocols; integrated **Confluence** documentation with **LIMS** systems to establish single source of truth for training and compliance.
- Automated routine laboratory administrative tasks—consumables ordering, stock maintenance, method verification scheduling, and EM (equipment monitoring) execution—using **Python** scripting and **Linux** cron jobs; coordinated with microbiology lab operations and internal audit functions to ensure change-control management and continuous improvement across interconnected systems.
- Supported site safety culture and cross-functional audit preparation by maintaining laboratory housekeeping documentation, identifying process improvement opportunities, and facilitating meetings between Enabling Technology, Analytical, and Operations teams; enabled timely team training completion and coaching on **LIMS**, electronic notebooks, and **MODA** platforms.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)